Nmap Host Discovery (Finding Online Hosts)

The first step in network scanning is to identify which hosts are online (alive).
Nmap uses several techniques to discover live hosts on a network.

The main option used for this is:

```
-sn
```

This is called a Ping Scan or Host Discovery Scan.
It discovers live hosts only and does not scan ports or services.

Example:

```
nmap -sn 192.168.1.0/24
```

Ways to Specify Targets in Nmap

Nmap allows multiple ways to define scan targets.

1. IP Range (-)

Used to scan a specific range of IP addresses.

Example:
192.168.0.1-10

Scans:
192.168.0.1 → 192.168.0.10

2. Subnet (/)

Used to scan an entire subnet.

Example:
192.168.0.1/24

Equivalent to scanning:
192.168.0.0 – 192.168.0.255

Total addresses:
256 hosts

3. Hostname

You can also scan using a hostname.

Example:
example.thm

Scanning a Local Network

A local network means a network directly connected to your device via:

- Ethernet
- WiFi

Example network:
192.168.66.0/24

Command:

```
nmap -sn 192.168.66.0/24
```

```
root@tryhackme:~# nmap -sn 192.168.66.0/24
Starting Nmap 7.92 ( https://nmap.org ) at 2024-08-07 13:49 EEST
Nmap scan report for XiaoQiang (192.168.66.1)
Host is up (0.0069s latency).
MAC Address: 44:DF:65:D8:FE:6C (Unknown)
Nmap scan report for S190023240007 (192.168.66.88)
Host is up (0.090s latency).
MAC Address: 7C:DF:A1:D3:8C:5C (Espressif)
Nmap scan report for wlan0 (192.168.66.97)
Host is up (0.20s latency).
MAC Address: 10:D5:61:E2:18:E6 (Tuya Smart)
Nmap scan report for 192.168.66.179
Host is up (0.10s latency).
MAC Address: E4:AA:EC:8F:88:C9 (Tianjin Hualai Technology)
[...]
Nmap done: 256 IP addresses (7 hosts up) scanned in 2.64 seconds
```

How Nmap Detects Local Hosts

In a local network, Nmap sends:

- ARP Requests
- If a device responds →
  - Nmap marks:
    - Host is up

Scanning a Remote Network

A remote network is a network separated by one or more routers.

Example:
Your network: 192.168.66.0/24
Target network: 192.168.11.0/24

Command:

```
nmap -sn 192.168.11.0/24
```

List Scan (-sL)

Nmap can also list targets without scanning them.

Command:

```
nmap -sL 192.168.0.1/24
```

Purpose:

- Shows which IPs will be scanned
- Does not send packets
- Useful for verification before scanning.

Discovering Network Services with Nmap

Each protocol supports 65,535 ports. So scanning tools like Nmap help determine which ports are open and running services.

TCP Port Scanning

To detect open ports, scanners attempt to communicate with each port.
If a port responds correctly, it means a service is listening.

Example idea using Telnet:

```
telnet target_ip port
```

This tries to establish a TCP connection.

If successful then the port is open.

TCP Connect Scan

A TCP Connect Scan attempts to fully establish a TCP connection.

Nmap option:

```
-sT
```

Example:

```
nmap -sT 192.168.1.10
```

It performs the TCP three-way handshake:

1. Client → SYN
2. Server → SYN-ACK
3. Client → ACK

If the handshake succeeds that means the port is open.

After connecting, Nmap immediately closes the connection.

SYN Scan (Stealth Scan)

Another common scan is the SYN Scan.

Option:

```
-sS
```

Example:

```
nmap -sS 192.168.1.10
```

1. Only the first step of the handshake is sent:

```
Client → SYN
```

2. If the port is open:

```
Server → SYN-ACK
```

3. Then Nmap immediately sends:

```
RST
```

This terminates the connection without completing the handshake.

It Is Called Stealth Scan Because:

- Connection is never fully established
- Fewer logs may be generated
- Harder for systems to detect

Requires root privileges.

UDP Port Scanning

Not all services use TCP. Many important services use UDP.

Examples:
DNS
DHCP
NTP
SNMP
VoIP

UDP scanning option:

```
-sU
```

Example:

```
nmap -sU 192.168.1.10
```

UDP Scan Behavior

- UDP has no handshake.
- So Nmap sends a UDP packet.
- If the port is closed, the host usually replies with:
    - ICMP Destination Unreachable (Port Unreachable)

If no response → port might be open or filtered.

UDP scanning is slower and less reliable than TCP scanning.

Limiting Target Ports

By default, Nmap scans the 1000 most common ports.
But we can control the number of ports scanned.

1. Fast Scan

Option:

```
-F
```

Scans only 100 common ports.

Example:

```
nmap -F 192.168.1.10
```

2. Specify Port Range

Option:

```
-p
```

Examples:

Scan ports 10 to 1024

```
nmap -p10-1024 192.168.1.10
```

Scan ports 1 to 25

```
nmap -p1-25 192.168.1.10
```

Scan all ports

```
nmap -p- 192.168.1.10
```

Summary of Important Nmap Options

Option	Meaning
-sT	TCP Connect Scan (full handshake)
-sS	TCP SYN Scan (stealth scan)
-sU	UDP Scan
-F	Fast scan (100 ports)
-p[range]	Scan specific ports
-p-	Scan all ports

Example: Finding a Web Server with Nmap

Sometimes web servers do not run on the default port (80). They may run on alternative ports such as:

- 8000
- 8008
- 8080
- 3000

To find them, we scan the target machine.

Example command using Nmap:

```
nmap -F 10.49.156.120
```

It scans the 100 most common ports, making the scan much faster.

Example Scan Result

PORT	STATE	SERVICE
7/tcp	open	echo
9/tcp	open	discard
13/tcp	open	daytime
22/tcp	open	ssh
8008/tcp	open	http

This means:
Port 8008 is open

It is running an HTTP (web server) service

Accessing the Web Server

Since the web server is on port 8008, we must include the port in the URL.

Open a browser and visit:

```
http://10.49.156.120:8008
```

The format is:
http://IP_ADDRESS:PORT

Version detection

OS Detection

Nmap can guess the operating system of a target using:

```
-O
```

Example:

```
nmap -sS -O 192.168.1.10
```

How OS Detection Works

- Nmap analyzes:
  - TCP/IP stack behavior
  - Packet responses
  - Network characteristics
- Then it makes an educated guess about the OS.

Example Output

```
Running: Linux 4.X|5.X
OS details: Linux 4.15 - 5.8
```

Service and Version Detection

To identify what service is running and its version, use:

```
-sV
```

Example:

```
nmap -sS -sV 192.168.1.10
```

Example Output

PORT	STATE	SERVICE	VERSION
22/tcp	open	ssh	OpenSSH 8.9p1 Ubuntu

Aggressive Scan (-A)

Instead of running multiple options separately, Nmap provides:

```
-A
```

Example:

```
nmap -A 192.168.1.10
```

-A Includes

- OS Detection (-O)
- Version Detection (-sV)
- Script Scanning
- Traceroute

When to Use -A

Use when you want:

- Maximum information quickly
- A full overview of the target

- Generates more network traffic (noisy)
- Easier to detect by IDS/IPS systems

Forcing the Scan (-Pn)

Normally, Nmap first checks if a host is alive using host discovery.

If the host does not respond, Nmap assumes:

The host is down and skips scanning.

Solution: Force Scan

Use:

```
-Pn
```

Example:

```
nmap -sS -Pn 192.168.1.10
```

-Pn Does

- Skips host discovery
- Treats all targets as online
- Forces port scanning

When to Use -Pn

- Target blocks ICMP (ping)
- Firewall is present
- Host appears down but is actually up

Summary of Important Nmap Options

Option	Meaning
-O	OS detection
-sV	Service & version detection
-A	Aggressive scan (OS + version + more)
-Pn	Skip host discovery (force scan)

Timing & Performance Control

When scanning with Nmap, controlling speed and timing is important for:

- Avoiding detection (IDS/IPS)
- Handling slow networks
- Optimizing scan performance

1. Timing Templates (-T)

Nmap provides 6 timing levels, from very slow (stealthy) to extremely fast (noisy).

Syntax

```
nmap -T<0-5> target_ip
```

or

```
nmap -T paranoid target_ip
```

Timing levels

Level	Name	Description
T0	Paranoid	Extremely slow, avoids detection
T1	Sneaky	Very slow, stealthy
T2	Polite	Slows down to reduce network load
T3	Normal	Default speed
T4	Aggressive	Faster, assumes good network
T5	Insane	Very fast, noisy, less reliable

Example

```
nmap -sS -F -T4 10.49.176.205
```

- -sS → SYN scan
- -F → Fast scan (100 ports)
- -T4 → Aggressive timing

Parallelism (How Many Probes at Once)

Nmap sends probes (requests) to ports.
You can control how many run at the same time.

Options

```
--min-parallelism <number>
--max-parallelism <number>
```

- Low value → slow but stable
- High value → fast but may miss data

Example

```
nmap --max-parallelism 50 target_ip
```

Nmap can send up to 50 probes at the same time

Rate Control (Packets per Second)

Controls how fast packets are sent.

Options

```
--min-rate <number>
--max-rate <number>
```

Example

```
nmap --min-rate 100 --max-rate 500 target_ip
```

Send between 100–500 packets per second

Higher rate → faster scan
Lower rate → stealthy scan

Host Timeout

Limits how long Nmap waits for a host.

Option

```
--host-timeout <time>
```

Example

```
nmap --host-timeout 30s target_ip
```

Stop scanning a host after 30 seconds

Summary

Option	Meaning
-T0 → -T5	Control scan speed
--min-parallelism	Minimum probes at once
--max-parallelism	Maximum probes at once
--min-rate	Minimum speed (packets/sec)
--max-rate	Maximum speed
--host-timeout	Max time per host

Verbosity & Saving Results

1. Verbosity (-v)

Sometimes scans take long and show very little output.

To see more real-time information, use:

```
nmap -v target_ip
```

Function of -v:

- Shows scan progress
- Displays what Nmap is doing step by step
- Helps you understand:
    - Host discovery

    - Port scanning
    - DNS resolution

Example

```
nmap -sS -v 192.168.1.0/24
```

```
-vv      (more details)
-vvv     (even more)
```

or:

```
-v2
-v4
```

2. Debugging Mode (-d)

If you want very detailed internal info, use debugging:

```
nmap -d target_ip
```

Levels

-d (basic debug)
-dd (more)
-d9 (maximum)

- Shows internal working of Nmap
- Produces a lot of output
- Mostly used for:
    - Troubleshooting
    - Advanced users

3. Saving Scan Results

You can save scan output in different formats.

1. Normal Output (Human-readable)

```
-oN filename
```

Example:

```
nmap -sS 192.168.1.1 -oN result.txt
```

2. XML Output

```
-oX filename
```

Example:

```
nmap -sS 192.168.1.1 -oX result.xml
```

Useful for:

- Automation
- Tools integration

3. Grepable Output

```
-oG filename
```

Example:

```
nmap -sS 192.168.1.1 -oG result.gnmap
```

Useful for:

- Searching with grep
- Parsing results

4. Save All Formats (Best Option)

```
-oA filename
```

Example:

```
nmap -sS 192.168.1.1 -oA scan
```

Output Files Created
scan.nmap → normal output
scan.xml → XML format
scan.gnmap → grepable format